

Manipal University Jaipur
Department of Data Science
Assignment-1: Information Retrieval [DS4147]

Date of Issue: 10/09/2025

Date of Submission: 15/09/2025

Note:

- **Submit Assignment on Time.**
 - **Submit Assignment on a Hard Register, No loose sheets allowed.**
 - **Submit Assignment at LHC-315D after 1:00PM before DoS.**
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1. Define the following terms with examples:

- Information Retrieval (IR)
- Precision vs. Recall
- Inverted Index

2. A query returns 12 documents, out of which 8 are relevant. If there are 20 relevant documents in the collection, calculate:

- Precision
- Recall

3. Given two documents represented as term vectors:

- $D1 = (2, 1, 0, 2)$
- $D2 = (1, 1, 1, 1)$

Compute the **Cosine Similarity** between D1 and D2. Interpret the result.

4. In an IR collection:

- Total documents = 1000
- 200 documents are relevant to query Q
- A term “retrieval” appears in 150 documents, out of which 100 are relevant.

a) Compute the probability that a document is relevant given it contains the term “retrieval.”

b) What does this probability tell us about the usefulness of the term?

5. Consider the following three documents:

- D1: "information retrieval system"
- D2: "retrieval model in IR"
- D3: "IR uses inverted index"

a) Construct an **Inverted Index** after removing stopwords.

b) Evaluate the Boolean query: **(retrieval AND model) OR index**.

c) Which documents are retrieved?

6. A query $Q = \{\text{retrieval, model}\}$ is given for a collection of 3 documents:

- D1: "information retrieval system"
- D2: "probabilistic model in retrieval"
- D3: "bm25 model for information retrieval"

Parameters:

- $N=3$, average doc length = 4
- Doc lengths: $D1=3$, $D2=4$, $D3=5$
- Term frequencies:
 - "retrieval": $D1=1$, $D2=1$, $D3=1$
 - "model": $D1=0$, $D2=1$, $D3=1$

a) Compute the **IDF** values for both query terms.

b) Using BM25 (with $k_1=1.2$, $b=0.75$, $k_1=1.2$, $b=0.75$), compute the BM25 score for each document.

c) Rank the documents.

7. Given the following term-document matrix for a corpus of 3 documents (terms already weighted using TF-IDF):

Term	D1	D2	D3
Retrieval	0.6	0.8	0.0
Model	0.4	0.6	0.7
System	0.9	0.0	0.3

a) Represent each document as a vector.

b) Compute the **Cosine Similarity** between:

- D1 and D2
- D2 and D3
- D1 and D3

c) Which pair of documents is the most similar? Justify.

8. Suppose we have a corpus of documents with a vocabulary of 5 terms:

["data", "mining", "retrieval", "information", "text"]

We consider:

- Document D1: "data mining and text"
- Document D2: "information retrieval and data"
- Query Q: "data retrieval"

Find Similarity of Q *w.r.t* the D1 & D2 using VSM.

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